

Software Upgrade Procedure for Waveserver 5

Software upgrade to OS 2.2.21

This Software Upgrade Procedure document supports upgrades to Waveserver 5 OS 2.2.21 from the following release:

- Waveserver 5 2.0.3, 2.0.4, 2.1.0, 2.1.1, 2.1.2, 2.2.0, 2.2.1 and 2.2.2

323-3001-900 - Issue 1

June 2023

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Publication History

Document release	Date (dd/mm/yyyy)	Engineer	Writer	Reason for change
Issue 1	16/06/2023	NM	NM	First release of the document

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Module 1 — Off-system preparation

Procedure 1

Prepare for the upgrade

Purpose

To:

- Obtain the OS 2.2.21 GA software load
- Obtain the latest issue of the Software Upgrade Procedure CTP 323-3001-900.
- Understand information relevant to the upgrade
- Ensure that Waveserver connectivity issues do not occur after the upgrade

Requirements

- Login credentials for <https://my.ciena.com>

Restrictions

- This document only covers software upgrades through CLI.
- Both single step and multi-step upgrade procedures are supported.
- Your network must support SSH2

Obtain OS 2.2.21 software load

- [] 1 Login to <https://my.ciena.com/> and download the OS 2.2.21 software load.

Obtain the documentation

- [] 2 Determine if you have the latest issue of this Software Upgrade Procedure (SUP) document by downloading the product documentation from the <https://my.ciena.com>.

[] 3	If	Then
	You have the latest issue of this SUP document	go to Step 6 (page 6)
	otherwise	go to Step 4 (page 6)

- [] 4 Download the latest issue of this SUP document from the Web site.

- [] 5 Replace your issue of this SUP document with the latest issue (save or print the document).

Know how to use this Software Upgrade Procedure document

- [] 6 Know that:
- this software upgrade procedure (SUP) document defines the “upgrade from software release” as the existing software release your system is running and the “upgrade to software release” as the new software release your system will be running after the SUP has been performed. In this SUP, the “upgrade to software release” is OS 2.2.21 and the “upgrade from software release” is:
 - Waveserver 5 2.0.3, 2.0.4, 2.1.0, 2.1.1, 2.1.2, 2.2.0, 2.2.1 and 2.2.2.
 - The software upgrade consists of activities documented in this SUP
 - This document details how to upgrade Waveserver 5 through CLI.
 - Ciena technical support contacts can be found in Table 2 (page 27)

Know the upgrade prerequisites

- [] 7 Before starting the upgrade you must have ordered and installed the OS 2.2.2 upgrade license. It is mandatory that no license-related alarms are raised on your system at the time of the upgrade, these will prevent the upgrade from initiating. Refer to the Software Licensing chapter of the appropriate release of the Waveserver 5 User Guide CTP 323-4002-100.



CAUTION

Upgrade Prerequisite

S86-LIC-WOSUPG020202 license must be ordered and installed prior to upgrading to OS 2.2.21.

- [] 8 Know the upgrade restrictions
- [] 9 For general upgrade restrictions, know that:



CAUTION

Supported Upgrade Paths

Only the upgrade paths listed below are supported using this document:

Waveserver 5 2.0.3, 2.0.4, 2.1.0, 2.1.1, 2.1.2, 2.2.0, 2.2.1 and 2.2.2 to OS 2.2.21.



CAUTION

Upgrade remotely managed nodes first

Upgrade nodes managed via remote comms before upgrading nodes with direct DCN access. After remote node is upgraded, remote comms will be temporarily unavailable until local node is also upgraded to OS 2.2.21.



CAUTION

Take note of NTP servers

Prior to upgrading to OS 2.2.21, take a note of all provisioned NTP servers:

```
# ntp client show
```

Once the upgrade is complete, compare the list of provisioned NTP servers, if any are missing, re-provisioned them:

```
# ntp client add server <IP address or host name>
```

**CAUTION****Maximum number of local users**

Prior to the upgrade make sure the number of local users does not exceed 50, using the following CLI command:

```
# user show
```

If there are more than 50 users, delete the excess using the following CLI command:

```
# user delete user <username>
```

**CAUTION****Check User Gateway settings**

If upgrading from OS 2.0.x, prior to the upgrade, check the local interface to ensure there's no user configured gateway provisioned:

```
# interface show
```

If BGP is enabled and IPv4 Gateway(User) is provisioned, remove it:

```
# interface unset interface local gateway
```

Otherwise make sure that the IPv4 Gateway(User) and IPv4 Gateway(Oper) values match.

If the values are different use the following command to provision the correct Gateway:

```
# interface set interface local gateway <IP_address>
```

**CAUTION****Check port and PTP status**

If upgrading from OS 2.0.x, prior to the upgrade, check the administrative state of the ports and PTPs for the facilities that are running traffic:

```
# port show
```

```
# ptp show
```

If any are unexpectedly disabled, enable them using the following command:

```
# port enable port <slot>/<port>
```

```
# ptp enable ptp <slot>/<port>
```

**CAUTION****gRPC service with mutual authentication**

If upgrading from OS 2.0.x with the gRPC server provisioned for mutual authentication, please make sure to re-install the certificate post upgrade.

1. Upload the gRPC server certificate to the node.

2. Install the certificate:

```
# certificates entity install cert-name <cert_name>  
filename /<dir>/<cert_name>.p12 cert-passphrase ****
```

3. Set gRPC server certificate name:

```
# system server grpc set cert-name <cert_name>
```

4. Enable mutual authentication:

```
# system server grpc mutual-authentication enable
```

Procedure 2

Gather and record the upgrade information

Purpose

To:

- gather and record the information you need for this upgrade
- perform configuration backup
- determine your requirements to perform each procedure

Requirements

An up-to-date map of your system.

Restrictions

No restrictions.

Record your name and the date

- [] 1 Enter your name and the date into Table 1 (page 12).

Record your node data

- [] 2 Use an up-to-date map of your system to determine the Node parameters and IP addresses of the nodes in your system to upgrade. In Table 1 (page 12), record the Waveserver 5 Parameters and IP address of all the nodes you need to upgrade.

Login to each node to be upgraded

- [] 3 Login to each node listed in Table 1 (page 12) using ssh and go to Step 4 (page 10)

Backup all the nodes



CAUTION

Stop all provisioning activities

If provisioning changes are made on the nodes after the backup is performed and prior to the upgrade, perform backup of those nodes again to ensure you have the latest copy of the configuration.

- [] 4 Stop all provisioning activities on the nodes to be upgraded. This includes disabling any active loopback(s) or Integrated Test Set(s).

- [] 5 For every node to be upgraded, check the active release.

`# software show`

If the active release is

Then

OS 2.0.x

Go to Step 6 (page 11)

OS 2.1.x

Go to Step 8 (page 11)

- [] 6 For every node to be upgraded, check default load and save filename

configuration list

If the default load name and the save name are

Then

different

Go to Step 7 (page 11)

the same

Go to Step 8 (page 11)

- [] 7 Set the Default Load Filename to be the same as the Default Save Filename:

configuration set default-load-filename <default-save-filename>

- [] 8 For every node to be upgraded, perform the following:

Save the running configuration:

configuration save

Perform backup:

configuration backup filename <string>

- [] 9 Transfer the configuration files from each node to an external xFTP server:

*# system xftp putfile local-filename <string> sftp-server
<ip_address> login-id <string> password <string>*

Record the node upgrade method

- [] 10 Check off the node upgrade method in the “Node upgrade method” field of Table 1 (page 12).

Record the software delivery method

In Table 1 (page 12), record the IP address of the xFTP server workstation in the “Upgrade to software release location” “IP address:” field.

- [] 11 Go to Procedure 3 (page 14)

Table 1
Upgrade information

Name:						Date:					
"Upgrade to" software release load name:						OS 2.2.21					
Node upgrade method						Single step: ☐ Multi-step: ☐					
"Upgrade to" software release location						xFTP server IP addr.:					
Path for the "upgrade to" software release:											
Procedure time estimate											
Node	Node Information				Cluster if applicable Primary / Member	IP address	Mngt Remote	SSL	SSH	SW Upgrade Completed	Comments
	Site Name	Site ID	Group ID	Member ID							
Node 1							☐	☐	☐	☐	
Node 2							☐	☐	☐	☐	
Node 3							☐	☐	☐	☐	
Node 4							☐	☐	☐	☐	
Node 5							☐	☐	☐	☐	
Node 6							☐	☐	☐	☐	
Node 7							☐	☐	☐	☐	
Node 8							☐	☐	☐	☐	
Node 9							☐	☐	☐	☐	
Node 10							☐	☐	☐	☐	
Node 11							☐	☐	☐	☐	
Node 12							☐	☐	☐	☐	
Node 13							☐	☐	☐	☐	
Node 14							☐	☐	☐	☐	
Node 15							☐	☐	☐	☐	
Node 16							☐	☐	☐	☐	
Node 17							☐	☐	☐	☐	
Node 18							☐	☐	☐	☐	
Node 19							☐	☐	☐	☐	
Node 20							☐	☐	☐	☐	
Node 21							☐	☐	☐	☐	
Node 22							☐	☐	☐	☐	
Node 23							☐	☐	☐	☐	
Node 24							☐	☐	☐	☐	
Node 25							☐	☐	☐	☐	
Node 26							☐	☐	☐	☐	
Node 27							☐	☐	☐	☐	
Node 28							☐	☐	☐	☐	
Node 29							☐	☐	☐	☐	
Node 30							☐	☐	☐	☐	
Node 31							☐	☐	☐	☐	
Node 32							☐	☐	☐	☐	

¹Note: This table is only an example of how the node information required for the upgraded can be organized.

Module 2 — On-system preparation

Procedure 3

Store new software load on xFTP server

Purpose	To store the new software load on xFTP server.
Requirements	The xFTP server should be accessible by the node(s).
Restrictions	None.

Login to the xFTP server

- [] 1 Open the FTP client application on the PC that contains the “upgrade to” software downloaded from Step 1 (page 6)
- [] 2 Login to the xFTP server, using the IP address and the login credentials

Upload the new software load to the xFTP server

- [] 3 Navigate to the xFTP server folder where the OS 2.2.21 load will be uploaded, and record the file path in Table 1 (page 12).
- [] 4 Transfer the OS 2.2.21 load, *waveserver-2.2.21.1-GA.tar.gz*, to the selected folder.

Close the connection to the xFTP server

- [] 5 Close the xFTP client application

Determine the upgrade option

[] 6	If you want to perform	Then go to
	a single step system upgrade from CLI	Procedure 4 (page 16)
	a multi-step system upgrade from CLI	Procedure 5 (page 20)

Module 3 — Core upgrade activities using CLI

Procedure 4

Performing one-step system software upgrade from CLI

Purpose

To perform an immediate download, activation, and installation of upgrade system software.

The new software load is automatically committed after activation completes and Waveserver restarts.

Requirements

- The upgrade software has been downloaded from the Ciena portal and saved to the designated remote server.
- Enough free memory space exists on the Waveserver to support the new software load. To ensure this, follow the procedure starting from Step 8 (page 19) to delete all unnecessary software loads from the Waveserver
- To access the software server where the upgrade software is located, you must have available the IP address or host name of the server, along with the user name and password of the SFTP or SCP server.
- You are logged in to node using an account with at least admin privileges.

Restrictions

- No license missing alarms are raised on the node being upgraded
-

**ATTENTION**

Not a Service Affecting Procedure, except for 400GE traffic for upgrades from OS 2.0.3, 2.1.0 and 2.1.1.

The upgrade to OS 2.2.21 will not impact traffic, except for 400GE traffic when upgrading from OS 2.0.3, 2.1.0, 2.1.1 and 2.1.2.

**ATTENTION**

Modem Class Mismatch alarm may be raised

During the upgrade from OS 2.0.x to OS 2.2.21, a Modem Class Mismatch alarm may be raised while both ends of the service are running different releases.

**ATTENTION**

Transient alarms may be observed

Transient alarms may be raised and cleared during the upgrade. These alarms include:

- ILAN Link Down
- Credentials Warning
- Configuration Database Mismatch
- Loss of Adjacency

Note that there is no operational impact, only additional historical log entries.

Establish communication with your surveillance personnel

- [] 1 Establish communication with your surveillance personnel. Maintain communication during the upgrade to keep your surveillance personnel aware of alarms due to the upgrade.

Stop maintenance and provisioning

Ensure that all personnel are aware that there should be no provisioning changes before you complete the upgrade.

Select the Waveserver node to upgrade

- [] 2 Use Table 1 (page 12) to determine the next node to be upgraded.

Performing the one step system software upgrade

- [] 3 Use the following command to download, activate, and commit the upgrade software:

```
# software install url <String> [delete-from-load]
```

Note: A Warm restart of the node will be initiated shortly after this command.

Note: The target file should end with `.tar.gz`

Note: delete-from-load is a recommended option that will automatically delete the previous load during the upgrade.

Retrieving the upgrade status

- [] 4 Use the following command to verify that the Active Release Version is *waveserver-2.2.21.1-GA*.

Note: The upgrade takes about 10 minutes, so you can run the command below several times during this period and verify the upgrade progress. The upgrade is completed when the Upgrade State is “idle” and the committed release version is OS 2.2.21. Also, verify that all modules have auto-upgraded successfully, and are showing operational status “up”:

```
# software show upgrade-status
# module show
```



ATTENTION

Some event logs might be missing post upgrade

As part of upgrading from OS 2.0.3 or 2.0.4 to OS 2.2.21, some of the historical event logs may no longer be available.



CAUTION

In Case of Upgrade Failure

If the upgrade fails, call Ciena technical support - see page 27.

Cold restart on traffic modules carrying 400GE traffic - optional

Note that steps 5 to 7 are optional and are not required to be done as part of the upgrade.



ATTENTION

400GE clients could be susceptible to margin degradation over time which may cause traffic hits.

Customer equipment/switch/router may observe 400GE traffic hits or local faults due to high BER. This issue is resolved in OS 2.2.1 (and OS 2.0.4) however a cold restart is required on traffic modules carrying 400GE traffic for this fix to take effect. Note that if you're upgrading from OS 2.0.4 and have already cold restarted the affected modules, a subsequent cold restart is not required.

- [] 5 Take note of the modules on the chassis with 400GE clients.
- [] 6 Use the following CLI command to perform a cold restart on those modules:

```
# module restart-cold slot <slot>
```



CAUTION

This step is traffic affecting

A cold restart will impact all traffic on that module.

- [] 7 Use the following CLI command to verify that the status of the cold restart:

```
# module show slot <slot>
```

Delete unnecessary software load(s)

- [] 8 Use the following CLI command to view all software loads available on the node:
software show installed
- [] 9 Using the list from the previous step, determine the loads that are no longer required. It is recommended to keep only the current running load; therefore delete all software versions prior to the one currently running on your system. To determine the current load your system is running, use the following CLI command:
software show
- [] 10 For each software load to be removed, use the following command to delete the specified version of software:
software delete version <String: 1...31>
- [] 11 Repeat the above process for all nodes documented in Table 1 (page 12).

Procedure 5

Performing a multi-step system software upgrade from CLI

Purpose

Use this procedure to manually download, then activate and commit the upgrade software only when required.

Requirements

- The upgrade software has been downloaded from the Ciena portal and saved to the designated remote server.
- Enough free memory space exists on the node to support the new software load. To ensure this, follow the procedure starting from Step 8 (page 19) to delete all unnecessary software loads.
- To access the software server where the upgrade software is located, you must have available the IP address or host name of the server, along with the user name and password of the SFTP or SCP server.
- You are logged in the node using an account with at least admin privileges.



ATTENTION

Not a Service Affecting Procedure, except for 400GE traffic for upgrades from OS 2.0.3, 2.1.0 and 2.1.1.

The upgrade to OS 2.2.21 will not impact traffic, except for 400GE traffic when upgrading from OS 2.0.3, 2.1.0, 2.1.1 and 2.1.2.



ATTENTION

Modem Class Mismatch alarm may be raised

During the upgrade from OS 2.0.x to OS 2.2.21, a Modem Class Mismatch alarm may be raised while both ends of the service are running different releases.

Establish communication with your surveillance personnel

- 1 Establish communication with your surveillance personnel. Maintain communication during the upgrade to keep your surveillance personnel aware of alarms due to the upgrade.

Stop maintenance and provisioning

- 2 Ensure that all personnel are aware that there should be no provisioning changes before you complete the upgrade.

Select the node to upgrade

- 3 Use Table 1 (page 12) to determine the next node to be upgraded.

Downloading the software version to Waveserver

- [] 4 Download the upgrade software:

```
# software download url <String>
```

Example:

```
# software download url
sftp://tester@10.133.34.58/waveserver-2.2.21.1-GA.tar.gz password
***
```

Note: The target file should end with “.tar.gz”.

Software download can take several minutes to complete. To view the status of this process, use the command “software show upgrade-status”.

Option to back-out of the upgrade

[] 5	If you want to	Then go to
	continue with the activation of the software release	Step 6 (page 21)
	cancel the upgrade and go to the previous software release	Step 1 (page 26)

Activate the upgrade software:

- [] 6 Activate the new software version

```
# software activate version waveserver-2.2.21.1-GA [auto-commit]
[delete-from-load]
```

Note: The software activate will initiate a warm restart of the node causing a temporary loss of management access. Once management access is regained, the following command can be used to check on the upgrade status:

```
# software show upgrade-status
```

- [] 7 If auto-commit is specified as part of Step 6 (page 21) go to Step 10 (page 22).

Option to back-out of the upgrade

[] 8	If you want to	Then go to
	commit new software	Step 9 (page 21)
	cancel the upgrade and go to the previous software release	Step 1 (page 26)

Committing the new software version

- [] 9 Manually commit the new software

```
# software commit [delete-from-load]
```

Note: Optionally if the delete-from-load option is included in the command above, the previous software load will be automatically deleted upon the completion of the commit.

- [] 10 Use the following command to verify that the Active Release Version is *waveserver-2.2.21.1-GA*.

Note: The command below can be run several times during this period to verify the upgrade progress. The upgrade is completed when the Upgrade State is Commit Complete. Also verify that the modules have auto-upgraded successfully and are displaying an operational status state of “up”.

```
# software show upgrade-status
# module show
```

**ATTENTION****Some event logs might be missing post upgrade**

As part of upgrading from OS 2.0.3 or 2.0.4 to OS 2.2.21, some of the historical event logs may no longer be available.

**CAUTION****In Case of Upgrade Failure**

If the upgrade fails, call Ciena technical support - see page 27.

Cold restart on traffic modules carrying 400GE traffic - optional

Note that steps 11 to 13 are optional and are not required to be done as part of the upgrade.

**ATTENTION****400GE clients could be susceptible to margin degradation over time which may cause traffic hits.**

Customer equipment/switch/router may observe 400GE traffic hits or local faults due to high BER. This issue is resolved in OS 2.2.1 (and OS 2.0.4) however a cold restart is required on traffic modules carrying 400GE traffic for this fix to take effect. Note that if you're upgrading from OS 2.0.4 and have already cold restarted the affected modules, a subsequent cold restart is not required.

- [] 11 Take note of the modules on the chassis with 400GE clients.
- [] 12 Use the following CLI command to perform a cold restart on those modules:

```
# module restart-cold slot <slot>
```

**CAUTION****This step is traffic affecting**

A cold restart will impact all traffic on that module.

- [] 13 Use the following CLI command to verify that the status of the cold restart:

```
# module show slot <slot>
```

Delete unnecessary software load(s)

- [] 14 Use the following CLI command to view all software loads available on the node:
`# software show installed`
- [] 15 Using the list from the previous step, determine the loads that are no longer required. It is recommended to keep only the current running load, therefore delete all software versions prior to the one currently running on your system. To determine the current load your system is running, use the following CLI command:
`# software show`
- [] 16 For each software load to be removed, use the following command to delete the specified version of software:
`# software delete version <String: 1...31>`
- [] 17 Repeat the above process until all nodes documented in Table 1 (page 12).

Module 4 — Appendix

Procedure 6

Canceling the Software Upgrade from CLI

Purpose	• To cancel an initiated upgrade and return a node to its pre-upgrade state.
Requirements	• A node where download of a new load has completed, but software has not committed.
Restrictions	• None

Cancel the software upgrade:



CAUTION
Service Affecting Procedure!

Software Cancel will be a service affecting procedure once software activation has initiated.



CAUTION
Logs will be lost

Any logs collected while running the new release will be lost after issuing a Software Cancel. Collect logs, if required, prior to Software Cancel. These include command and event logs.

- [] 1 Cancel the software upgrade:

```
# software cancel
```

Note: A reboot will begin shortly after this command.

- [] 2 Once the node is reachable, wait for the traffic module(s) to initialize.

```
# module show
```

Note: The state of all modules should be “UP” before proceeding to the next step.

- [] 3 Cold restart the chassis:

```
# chassis restart-cold
```

Ciena technical assistance contacts

Table 2:
Technical assistance service telephone numbers

Corporate Headquarters	410-694-5700 or 800-921-1144	www.ciena.com
Customer Technical Support/Warranty		
In North America	1-800-CIENA-24 (243-6224) 410-865-4961	
In Europe, Middle East, and Africa	800-CIENA-24-7 (800-2436-2247) +44-207-012-5508 00 0800 77 454 (Slovenia)	
In Asia-Pacific	800-CIENA-24-7 (800-2436-2247) +81-3-6367-3989 +91-124-4340-600 120 11104 (Vietnam) 000 8004401369 (India)	
In Caribbean and Latin America	800-CIENA-24-7 (800-2436-2247) 1230-020-0845 (Chile) 009 800-2436-2247 (Colombia) 0800-77-454 (Mexico and Peru) 00 008000442510 (Panama)	
Sales and General Information	North America: 1-800-207-3714 International: +44 20 7012 5555	E-mail: sales@ciena.com
In North America	410-694-5700 or 800-207-3714	E-mail: sales@ciena.com
In Europe	+44-207-012-5500 (UK)	E-mail: sales@ciena.com
In Asia	+81-3-3248-4680 (Japan)	E-mail: sales@ciena.com
In India	+91-22-42419600	E-mail: sales@ciena.com
In Latin America	011-5255-1719-0220 (Mexico City)	E-mail: sales@ciena.com
Training		E-mail: learning@ciena.com

Waveserver 5

Software Upgrade Procedure

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OS 2.2.21

Publication: 323-3001-900

Document issue: Issue 1

Document release date: June 2023

CONTACT CIENA

For additional information, office locations, and phone numbers, please visit the Ciena web site at **www.ciena.com**

